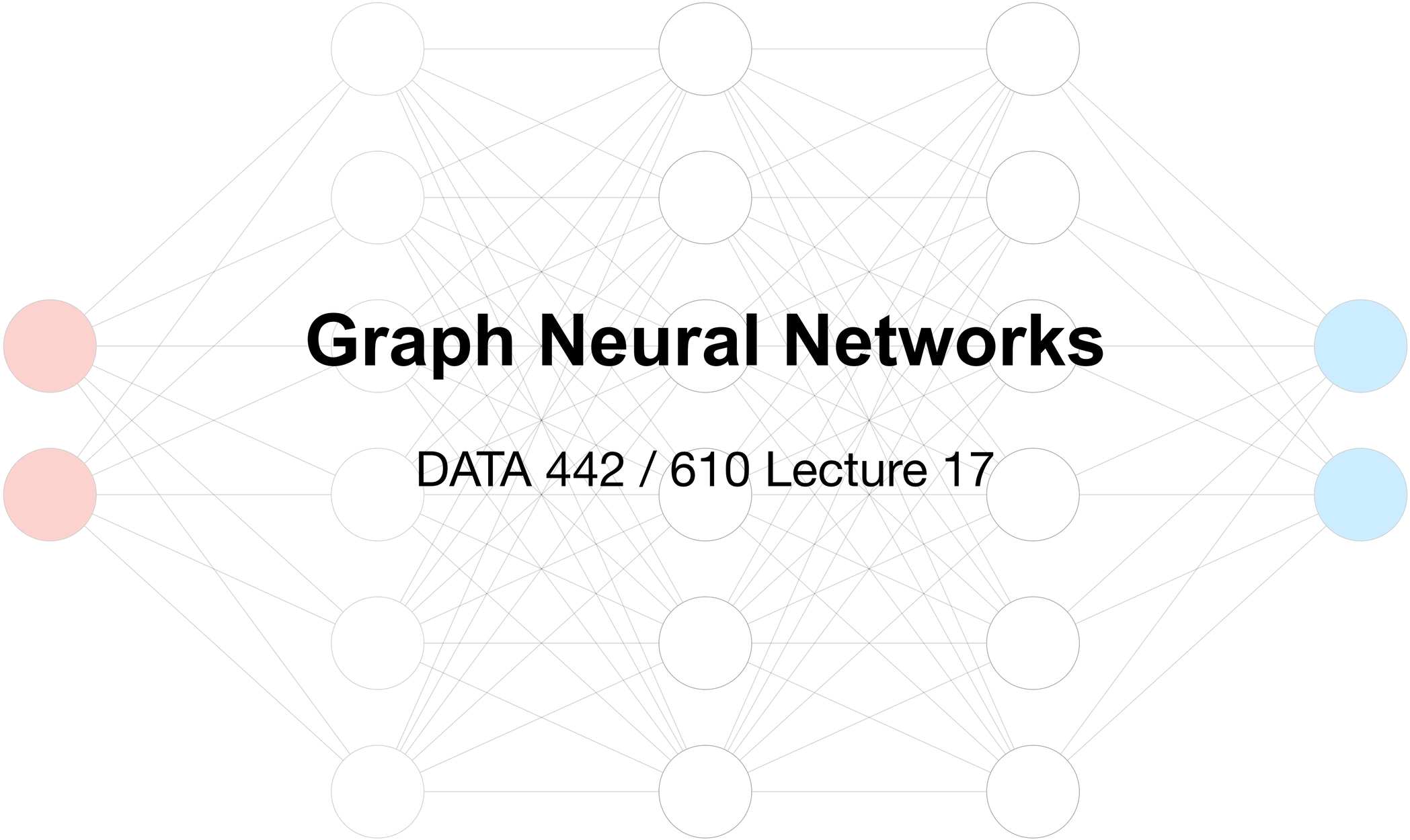


Graph Neural Networks

DATA 442 / 610 Lecture 17

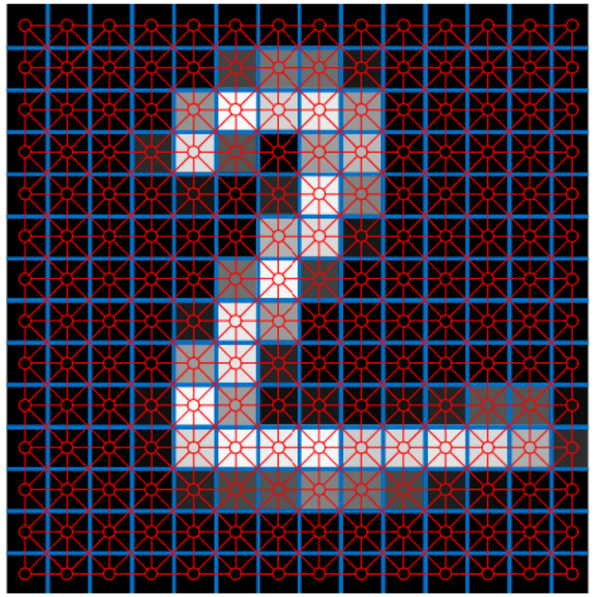
Outline

- Review of Graphs, Graph Convolutions
- Graph pooling and batch manipulation
- GNNs with Torch Geometric

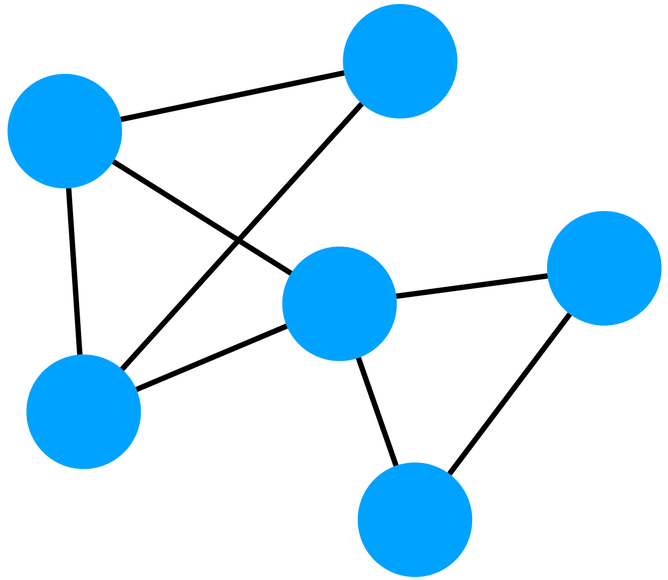
Relevant reading: ArXiv:2104.13478, distill.pub

Graphs

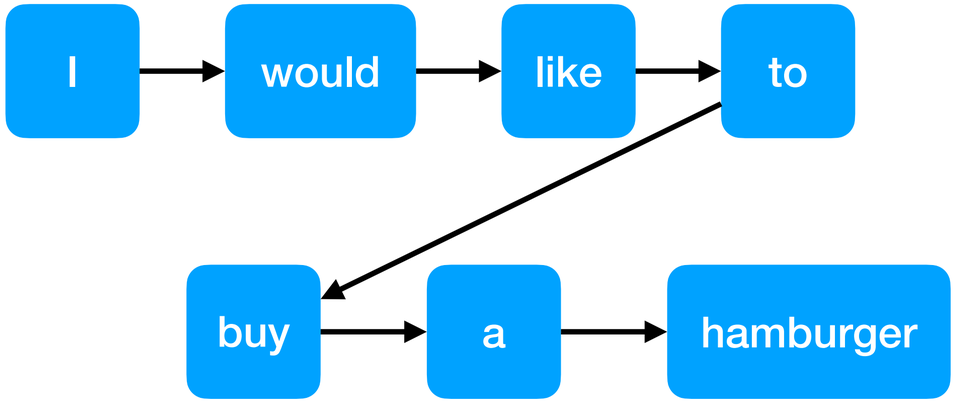
Image data



Graph data

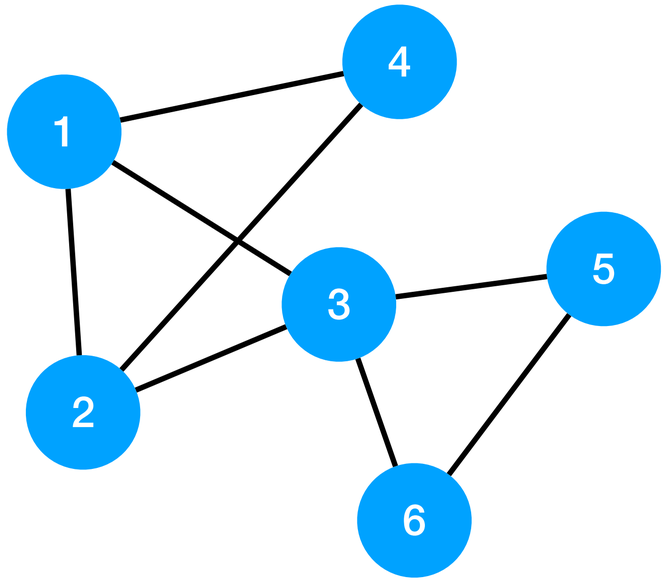


Sequential data



A **graph** is a structure containing **nodes** V and **edges** $E \subseteq V \times V$

Graphs



Adjacency matrix: $A_{ij} = \begin{cases} 1 & \text{if edge between } i \text{ and } j \\ 0 & \text{otherwise} \end{cases}$

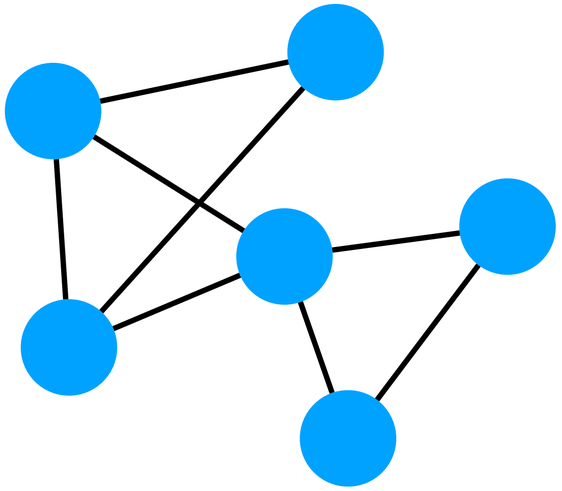
$$A = \begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Nodes: $V = \{1,2,3,4,5,6\}$

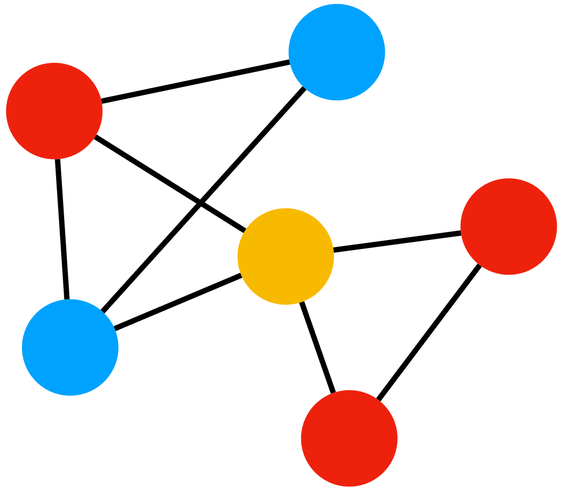
Edges: $E = \{(1,2), (1,3), (1,4), (2,3), (2,4), (3,5), (3,6), (5,6)\}$

Node features: $\mathbf{X} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \mathbf{x}_3 \\ \mathbf{x}_4 \\ \mathbf{x}_5 \\ \mathbf{x}_6 \end{pmatrix} = \begin{pmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,d} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,d} \\ x_{3,1} & x_{3,2} & \cdots & x_{3,d} \\ x_{4,1} & x_{4,2} & \cdots & x_{4,d} \\ x_{5,1} & x_{5,2} & \cdots & x_{5,d} \\ x_{6,1} & x_{6,2} & \cdots & x_{6,d} \end{pmatrix}$

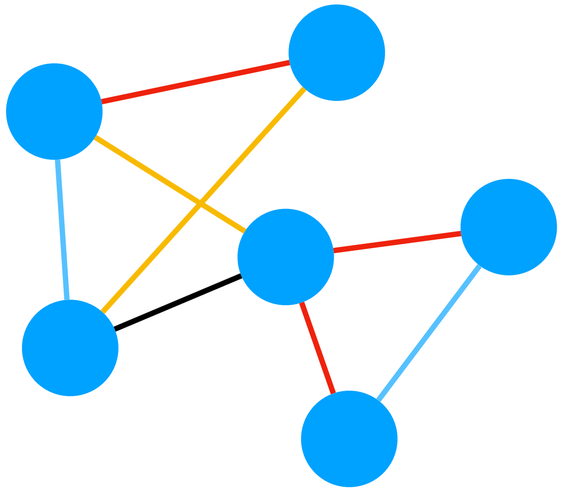
Graph problems



Node labeling



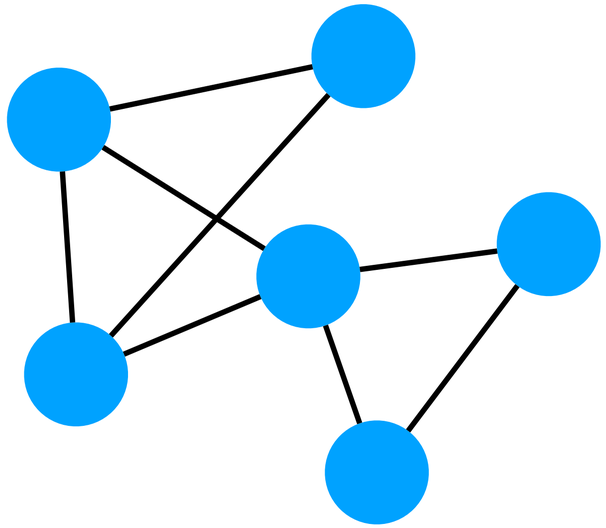
Edge labeling



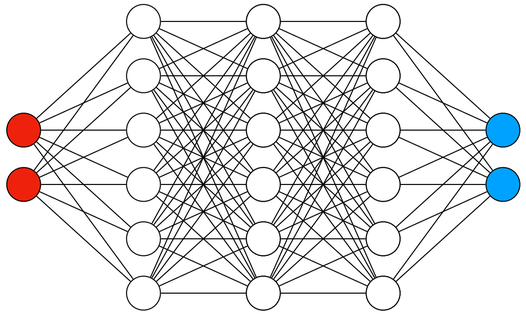
Graph-level labeling

“This is a car”

Deep learning on graphs



$$\mathbf{X} = \begin{pmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,d} \\ x_{2,1} & x_{2,2} & \dots & x_{2,d} \\ x_{3,1} & x_{3,2} & \dots & x_{3,d} \\ x_{4,1} & x_{4,2} & \dots & x_{4,d} \\ x_{5,1} & x_{5,2} & \dots & x_{5,d} \\ x_{6,1} & x_{6,2} & \dots & x_{6,d} \end{pmatrix}$$

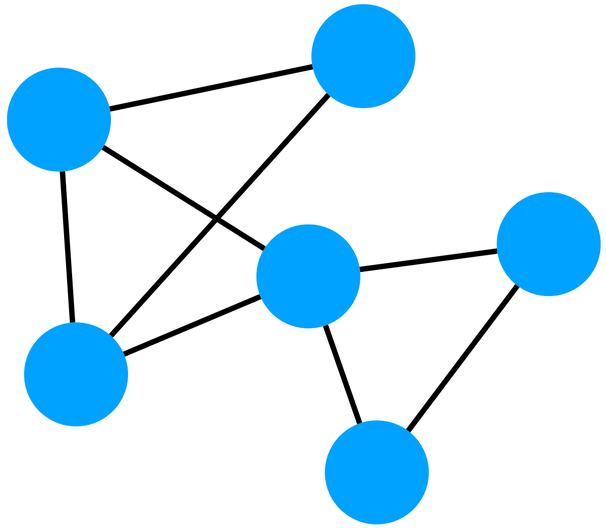


Graph-level labeling

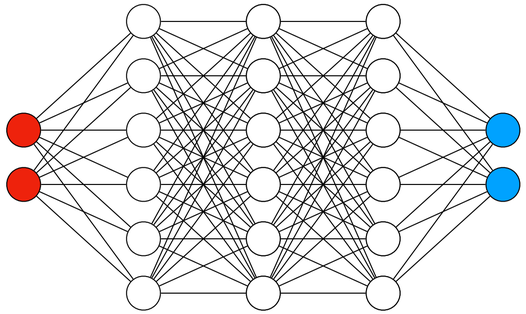
"This is a car"

Proposal: Process node features together using a neural network

Deep learning on graphs

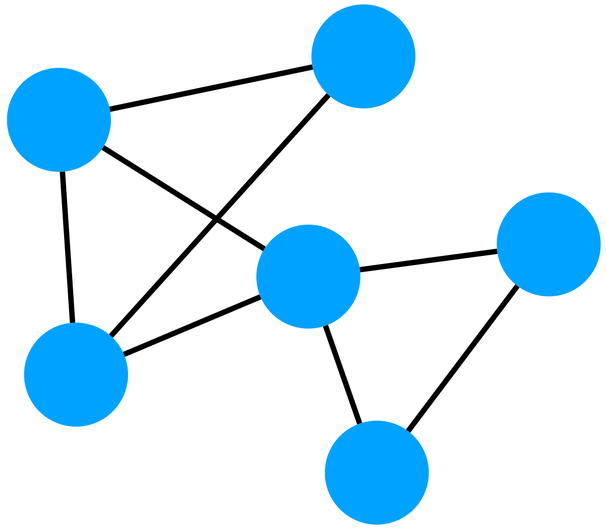


$$\mathbf{X} = \begin{pmatrix} x_{1,1} & x_{1,2} & \dots & x_{1,d} \\ x_{2,1} & x_{2,2} & \dots & x_{2,d} \\ x_{3,1} & x_{3,2} & \dots & x_{3,d} \\ x_{4,1} & x_{4,2} & \dots & x_{4,d} \\ x_{5,1} & x_{5,2} & \dots & x_{5,d} \\ x_{6,1} & x_{6,2} & \dots & x_{6,d} \end{pmatrix}$$

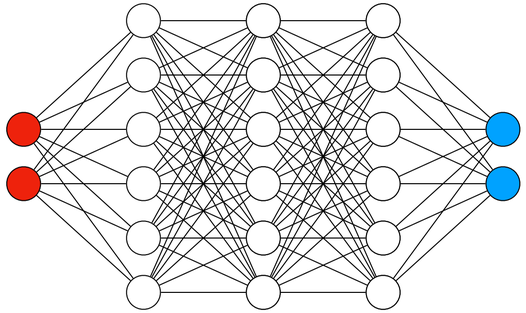


“This is a car”

Permutation P

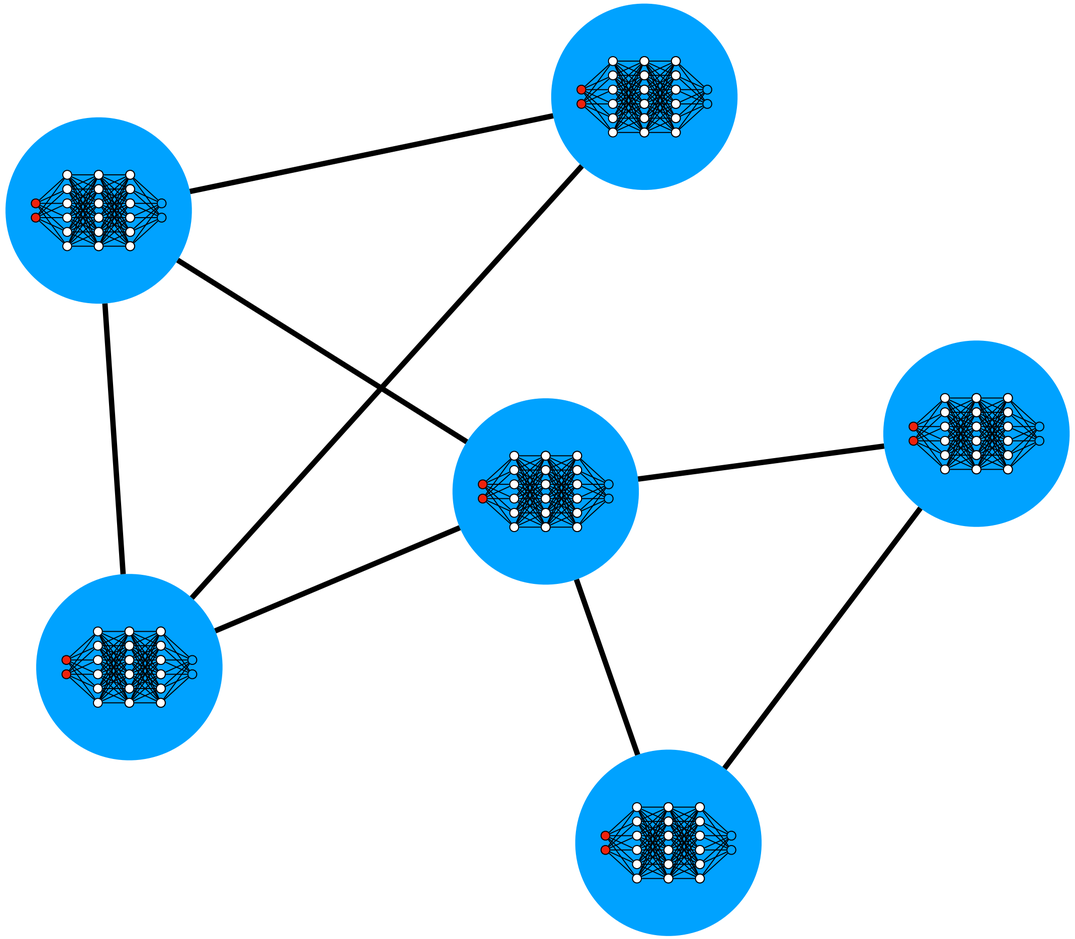


$$\mathbf{X} = \begin{pmatrix} x_{2,1} & x_{2,2} & \dots & x_{2,d} \\ x_{5,1} & x_{5,2} & \dots & x_{5,d} \\ x_{1,1} & x_{1,2} & \dots & x_{1,d} \\ x_{3,1} & x_{3,2} & \dots & x_{3,d} \\ x_{6,1} & x_{6,2} & \dots & x_{6,d} \\ x_{4,1} & x_{4,2} & \dots & x_{4,d} \end{pmatrix}$$



“This is a dog”

Deep learning on graphs

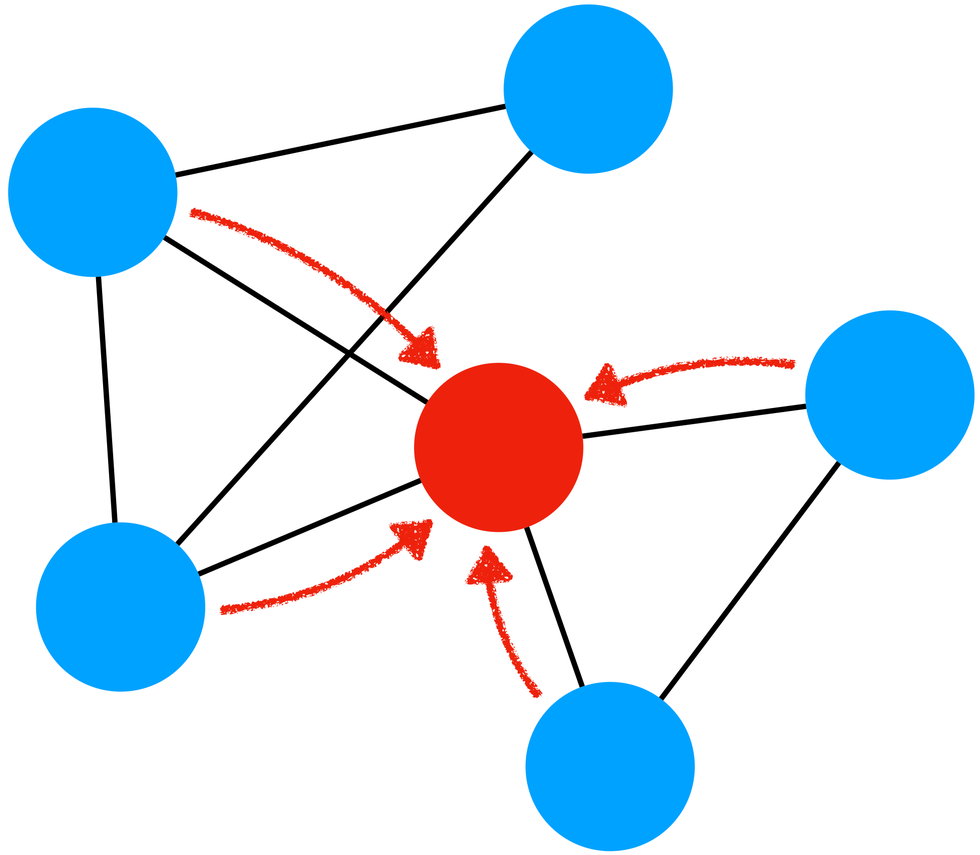


Proposal: process each node feature individually using deep NN

Requires a pooling operation

Does not exploit *relationships* between nodes

Graph convolution



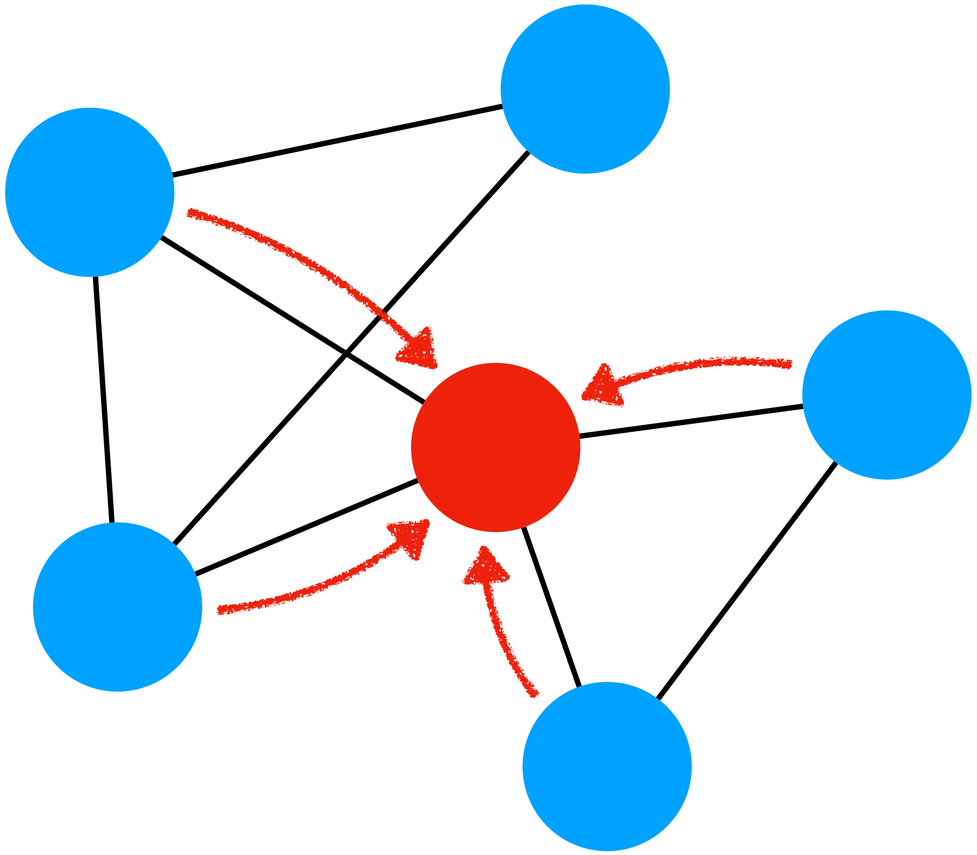
Graph convolution learns to update node features using neighbor information

$$\mathbf{x}_i^{\ell+1} = \sigma\Bigg(\underbrace{\mathbf{b}^\ell + \mathbf{x}_i^\ell \mathbf{W}_1^\ell}_{\text{self update}} + \underbrace{\sum_{j \in \mathcal{N}(i)} \mathbf{x}_j^\ell \mathbf{W}_2^\ell}_{\text{neighbor info}}\Bigg)$$

Requires adjacency matrix $\sum_{j \in \mathcal{N}(i)} \mathbf{x}_j = \sum_j A_{ij} \mathbf{x}_j$

$$\mathbf{X}^\ell = \mathbf{b}^\ell + \left(\mathbf{X}^\ell \mathbf{W}_1^\ell + A\,\mathbf{X}^\ell \mathbf{W}_2^\ell\right)$$

Graph convolution



Graph convolution operations generally involve two steps:

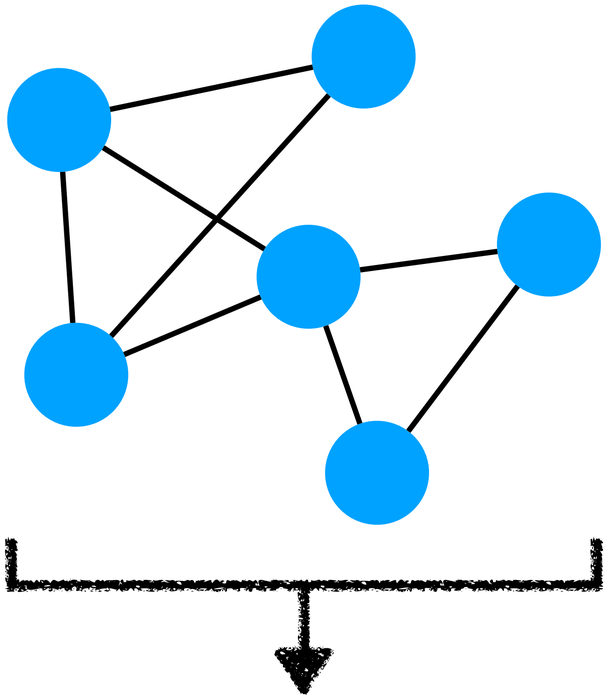
1. **Message passing:** nodes send and receive messages from their neighbors

$$\mathbf{m}_i = \sum_{j \in \mathcal{N}(i)} M(\mathbf{x}_i, \mathbf{x}_j, e_{ij})$$

2. **Node update:** nodes update their features based on incoming messages

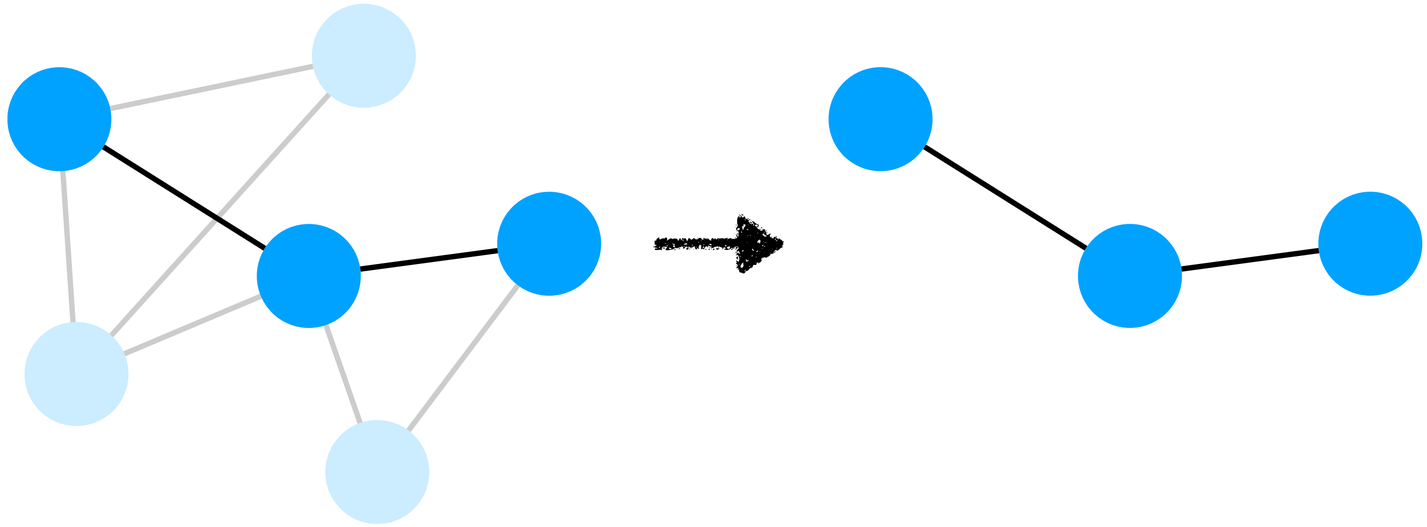
Graph pooling

Global pooling

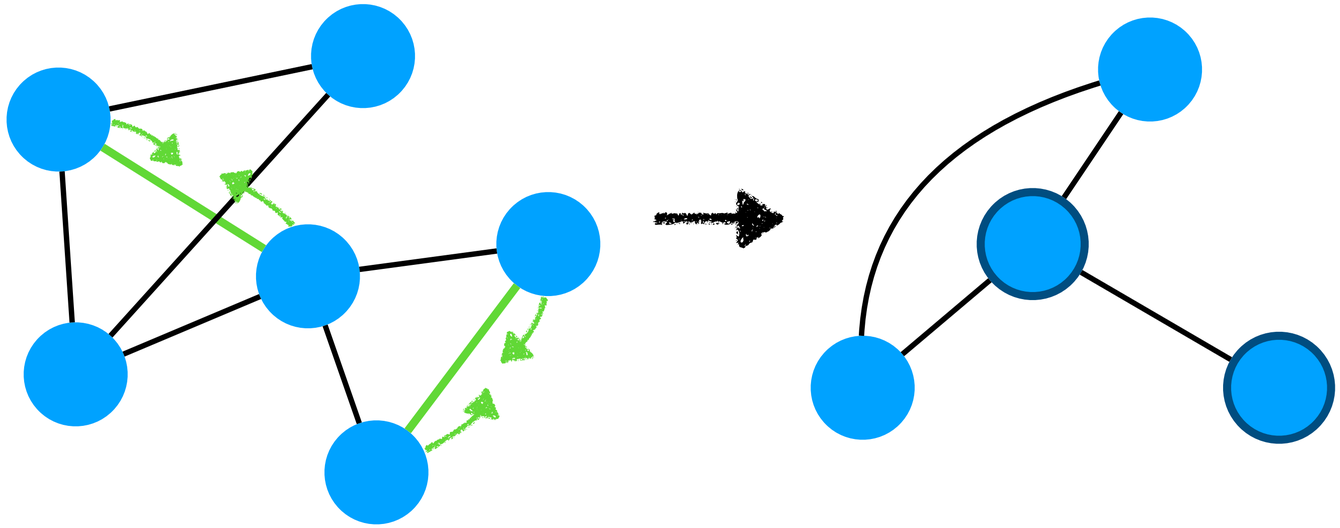


max, mean, sum, min, …

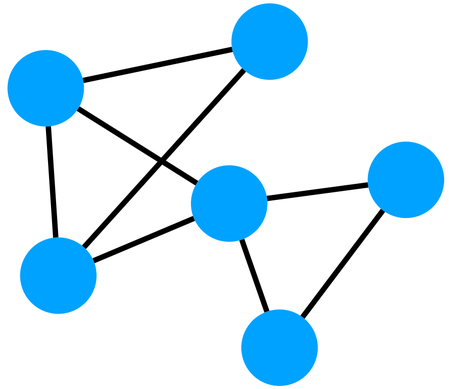
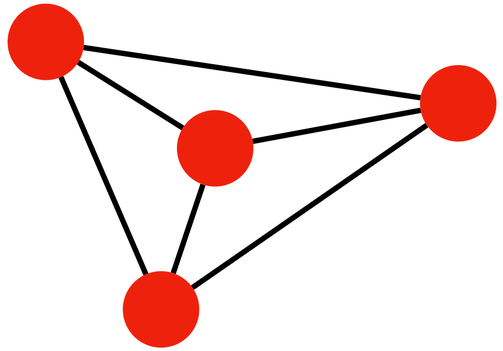
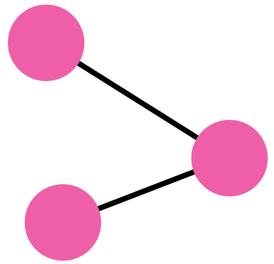
Local (top-k) pooling



Edge contraction



Graph batching

$$B = [$$  ,  ,  $$]$$

Batch node features

$$X_B = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} = \begin{pmatrix} x_{1,1,1} & \cdots & x_{1,1,d} \\ \vdots & \ddots & \vdots \\ x_{1,6,1} & \cdots & x_{1,6,d} \\ x_{2,1,1} & \cdots & x_{2,1,d} \\ \vdots & \ddots & \vdots \\ x_{2,4,1} & \cdots & x_{2,4,d} \\ x_{3,1,1} & \cdots & x_{3,1,d} \\ \vdots & \ddots & \vdots \\ x_{3,3,1} & \cdots & x_{3,3,d} \end{pmatrix}$$

Batch adjacency matrix

$$A_B = \begin{pmatrix} A_1 & 0 & 0 \\ 0 & A_2 & 0 \\ 0 & 0 & A_3 \end{pmatrix}$$

Batch feature computation

$$A_B X_B W_i = \begin{pmatrix} A_1 & 0 & 0 \\ 0 & A_2 & 0 \\ 0 & 0 & A_3 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \\ X_3 \end{pmatrix} W_i = \begin{pmatrix} A_1 X_1 W_i \\ A_2 X_2 W_i \\ A_3 X_3 W_i \end{pmatrix}$$

Batch selection mask

$$M_B = \begin{pmatrix} 1\,1\,1\,1\,1\,1 & 0 & 0 \\ 0 & 1\,1\,1\,1 & 0 \\ 0 & 0 & 1\,1\,1 \end{pmatrix}$$

Global sum pooling

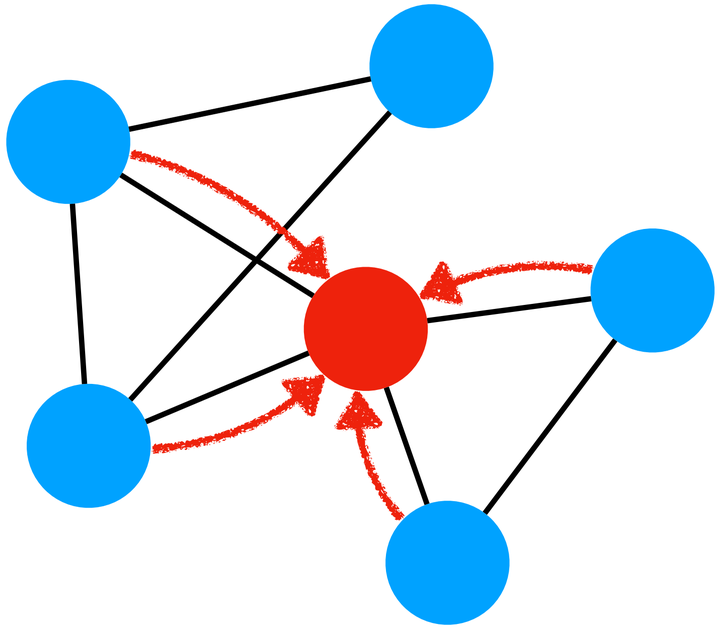
$$X_B M_B = \begin{pmatrix} \sum X_1 \\ \sum X_2 \\ \sum X_3 \end{pmatrix}$$

Hands-on demo for graph pooling and batching



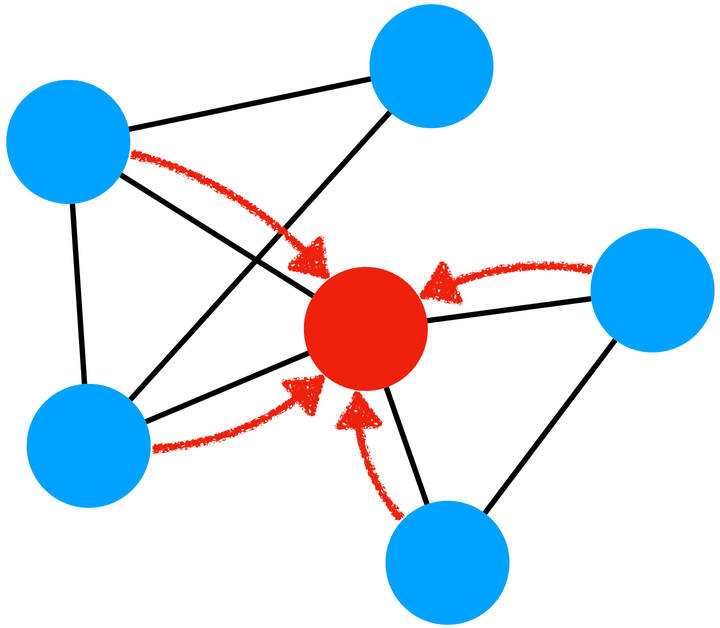
Graph operations

Convolutional

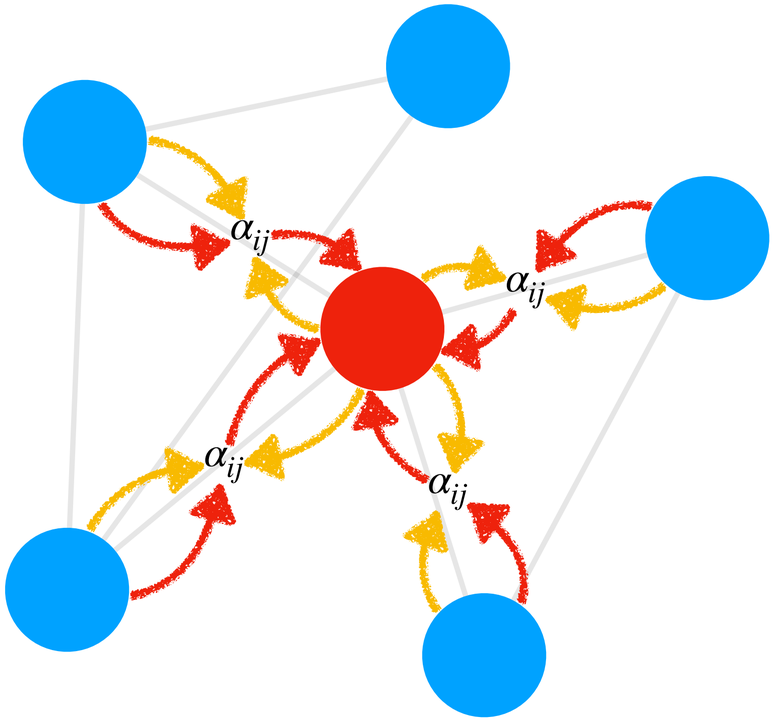


Graph operations

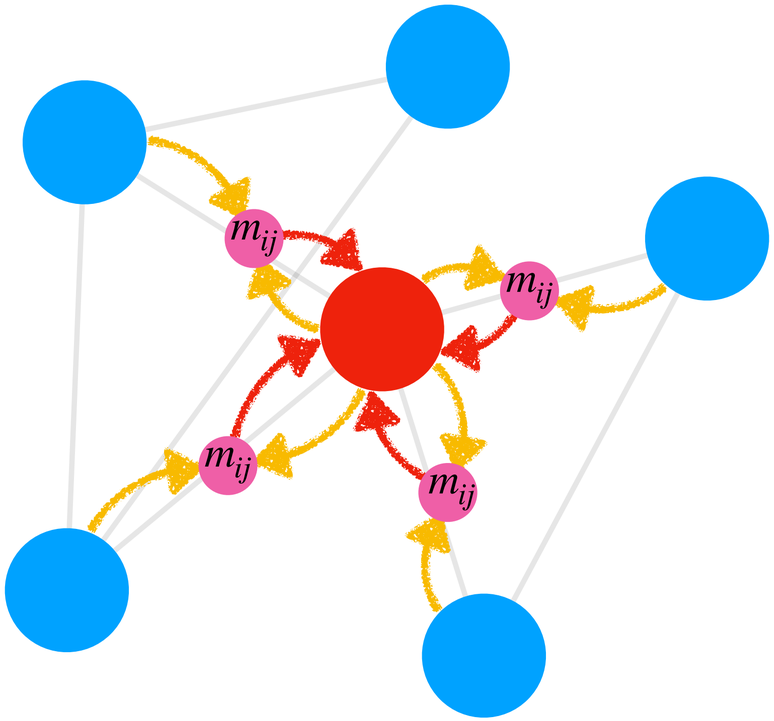
Convolutional



Attentional



Message passing



Hands-on demo with torch geometric



That’s all!

- Slides will be posted on the course website: https://colen-teaching.github.io/DATA442-Fall-2026
- Lab demos will also be posted
- ***PS3*** due November 8
- **REMEMBER:** TA Office Hours Tuesday 12-1pm, Student Success Center